



Assignment: K-Means Clustering Implementation

Dataset Options (Choose One)

You must train your clustering model on a standard dataset and then test with your own custom data. For Example:

1. Customer Segmentation: Train on a standard Mall Customers dataset (Age, Annual Income, Spending Score).

- **Your Custom Data:** Survey 10 individuals from the real world. Record their Age, estimated Annual Income, and a self-rated Spending Score (1-100).

2. Morphological Measurement (e.g., Iris or Palmer Penguins): Train on a standard botanical or biological dataset.

- **Your Custom Data:** Go outside and physically collect 10 distinct leaves or flowers. Measure their length and width in centimeters using a ruler.

3. Image Color Quantization: Train on a standard high-resolution image to cluster its RGB pixels into a reduced color palette.

- **Your Custom Data:** Take a photo with your smartphone. Use your trained model to compress the colors of your custom photo.

1. Build a Full Clustering Workflow

You must construct a comprehensive Machine Learning pipeline using Python and Scikit-Learn. Your pipeline must bridge standard dataset training with real-world testing.

- **Project on GitHub:** Storing code, custom data, and the trained model object (e.g., `.pkl` or `.joblib`).
- **Automatic data retrieval:** The notebook must automatically pull the standard dataset and your custom data directly from your GitHub repository.
- **Data Preprocessing:** Use `sklearn.preprocessing` for standardizing and scaling data.
- **Real-World Testing:** You must properly format your custom data, pass it through the pipeline and use your trained model to assign clusters.

2. Perform Essential Data Preprocessing & Custom Data Creation

Distance-based clustering requires strict data formatting. Your pipeline must include:

Standard Dataset Preprocessing:

- **Transforms:** K-Means relies heavily on Euclidean distance, therefore, u

